

Deepak Mallampati

APPLIED AI ENGINEER - FORWARD DEPLOYED

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
US-based, F-1 OPT - open to remote/relocation

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building and delivering production LLM and ML systems for banking, enterprise, and healthcare. Forward-deployed by instinct: I embed with teams, wire APIs into their pipelines, and turn requirements into working software fast - 12 APIs and microservices serving 100,000+ requests/day across 20+ teams at Vanguard, with p95 latency cut from 1.5s to 800ms and 99.9% uptime. Strong on Python/FastAPI integration, RAG and agentic systems, reliable automation (~60% failed runs), and observability - the glue work that makes customer integrations clean and durable.

CORE COMPETENCIES

Forward-Deployed / FDE Delivery • API Integration & SDKs • Python / FastAPI • REST Microservices • RAG & Agentic Systems • Data Pipelines • Observability & Evals • Docker / Kubernetes • Customer-Facing Engineering

Integration & Backend: Python, FastAPI, REST APIs, microservices, PostgreSQL, MongoDB, Redis, Kafka, JavaScript/TypeScript, React

Generative AI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LangChain/LangGraph, RAG, multi-agent systems, fine-tuning (LoRA/QLoRA), Hugging Face

Retrieval: FAISS, Pinecone, Weaviate, hybrid search, cross-encoder re-ranking, semantic caching

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda, S3), Azure OpenAI, GCP Vertex AI, Docker, Kubernetes, Terraform, CI/CD, MLflow, W&B, monitoring

Security & Compliance: HIPAA, SOC 2, PCI DSS, OAuth 2.0/JWT/RBAC, PII redaction, audit logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%** - the kind of resilient glue customer integrations depend on.
- Architected production **agentic LLM systems** on a conductor-subagent framework, decomposing objectives into context-scoped subagents and reducing token consumption by **42%** while sustaining accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large healthcare corpora (dense retrieval + cross-encoder re-ranking) to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions before release.
- Designed **privacy-preserving data workflows** (k-anonymity, l-diversity, differential privacy) for sensitive clinical data, enabling compliant analytics with zero PII exposure.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver LLM applications supporting **100,000+ requests daily** across 20+ internal teams - embedding with stakeholders to ship working integrations.
- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ agents** and workflows.
- Optimized prompts and evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates by **27%** and informing model-selection decisions.
- Engineered content safeguards for policy-compliant responses, reducing unsafe outputs by **42%** while maintaining quality.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline on a clinical readmission dataset, cutting record-linkage re-identification risk from **3.38% to 0.32%** while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 over a curated instruction dataset for controllable long-form generation.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Trained regression/classification models for equipment-failure prediction and anomaly detection; built end-to-end model-ready data pipelines.
- Fine-tuned BERT/RoBERTa for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga/webtoons across 29+ sites - a single inline action replacing a multi-step manual chore.

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - end-to-end integration of scanning, scoring, and document generation.

EDUCATION

M.S. in Computer Science, Kent State University, OH

2025 | GPA 3.70/4.0